Sahaj Bhadja

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Summary

Software engineer with 2+ years improving **product quality** through **automated testing**, **defect isolation**, and **reliable release workflows** across full-stack and cloud-based applications.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*

- Implemented **validation and reconciliation checks** across 5K+ monthly FastAPI jobs, improving production stability and reducing downstream defects in data-driven workflows.
- Diagnosed and resolved data reliability issues via SQL/query tuning and structured debugging, reducing latency from **1.2s to 350ms** for 500+ end users.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*

- Built and tested **React + Flask** grading workflows processing 10K+ daily requests, validating API outputs and edge cases for text/image submissions.
- Strengthened release quality with **CI/CD gates** and regression checks, reducing turnaround time by **60%** while maintaining consistent user-facing behavior.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*

- Developed and maintained **20+ unit/integration test suites** with PyTest and Mockito, increasing pipeline efficiency by **68%** and improving defect detection coverage.
- Used Jenkins-based automation and issue triage to reduce technical debt by **20%**, improving release readiness across Windows, Linux, and macOS environments.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*

- Automated **deployment and test execution** with Docker and Jenkins, cutting deployment time by **50%** and improving consistency of production rollouts.
- Built REST/WebSocket modules for eight workflows and 2M+ synchronized records, then validated behavior under concurrent usage to ensure stable operation.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*

- Created reproducible **testing pipelines** over 100K+ trajectory logs to evaluate model behavior, isolate failure modes, and support high-confidence implementation decisions.
- Produced defect-oriented diagnostics and technical summaries that improved cross-team communication on bug root causes and remediation priorities.

Skills

Testing/QA: **Unit & Integration Testing**, defect triage, regression checks, release validation
Tools: **Jenkins**, **PyTest**, Mockito, Git, CI/CD pipelines
Languages: Python, JavaScript/TypeScript, Java, SQL
Systems: Flask, FastAPI, REST APIs, Docker, AWS, Azure SQL, Linux/Windows

Projects

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Built schema-aware SQL generation with **query validation** and structured outputs across 10+ databases, improving correctness and reducing manual debugging effort.

Linguistix | *Python, FastAPI, AWS ECS/Fargate, CodePipeline*

- Developed a containerized REST API with **automated test and deployment checks**, achieving **87%** processing accuracy with stable cloud release workflows.

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB*

- Implemented secure full-stack transaction workflows and validated real-time sync paths for 50K+ users, supporting reliable behavior under high-volume usage patterns.